Aidan Syrgak

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EDUCATION

Northeastern University, Master of Science in Electrical and Computer Engineering | Boston, MA — May 2027
Worcester Polytechnic Institute, Bachelor of Science in Data Science | Worcester, MA — Dec 2024

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, R, SQL, HTML, CSS
Libraries: Pandas, NumPy, Matplotlib, SciPy, Selenium, TensorFlow, PyTorch, Scikit-learn
Frameworks and Tools: Apache Hadoop, Spark, PostgreSQL, Power BI, Tableau, Django, Flask

EXPERIENCE

Augmented Cognition Lab - Northeastern University — Boston, MA
Research Assistant — Jan 2026 - Present

- Analyzed brain-signal data from cognitive task experiments to identify patterns in neural responses
- Developed machine learning pipelines for neural response classification, evaluating diverse signal processing techniques and model architectures

Orsted North America — Boston, MA
Data and Permitting Intern — Jun - Aug 2024

- Engineered datasets of usable permit data by consolidating and cleaning information from the company's various renewable energy projects
- Developed a Power BI dashboard tool, enhancing data visualization and automation for the Permitting Team

Worcester Polytechnic Institute — Worcester, MA
Office Assistant — Nov 2023 - Apr 2024

- Developed Python scripts to process enrollment data, generate reports, and automate tasks for the Mathematical Sciences department
- Oversaw course scheduling, tutoring space readiness, and visitor assistance to ensure smooth operations

Reading Municipal Light Department — Reading, MA
Data Analyst Intern — May - Aug 2023

- Built Pandas-based pipelines to clean, transform, and analyze customer electricity meter data
- Designed interactive Power BI dashboards for the Integrated Resources Department, covering key areas of energy usage, electric vehicle impacts, and renewable technology

PROJECTS

Secure Voice-Controlled Car Door Model — Boston, MA
Northeastern University — Sep - Dec 2025
A voice-controlled car door system that is secure against electromagnetic, ultrasound, and laser injection attacks

- Designed a multi-layer security architecture on Raspberry Pi using dual-mic energy analysis, light sensors, and FFT-based spectral analysis
- Implemented real-time audio pipeline with VAD triggering, dual-channel recording, Goertzel-based DTMF decoding, and Google STT for command recognition

Hospital Navigation and Service Request Application — Worcester, MA
Worcester Polytechnic Institute — Jan - Mar 2023
A hospital operations app for Brigham and Women's Hospital

- In a ten-person team, applied Agile methodologies to build an indoor navigation and facilities management app, showcasing potential features and design approaches for hospital consideration
- Contributed to the development of the database using PostgreSQL and the backend in Java